

# Claude Cournoyer-Cloutier

✉ [cournoyc@MPA-Garching.MPG.DE](mailto:cournoyc@MPA-Garching.MPG.DE) | 🏠 <https://cournoyercloutierc.github.io/> | 🆔 0000-0002-6116-1014

## Employment

|                                                                           |                                    |
|---------------------------------------------------------------------------|------------------------------------|
| <b>Postdoctoral Fellow</b><br>MAX PLANCK INSTITUTE FOR ASTROPHYSICS (MPA) | Garching, Germany<br>Since 10/2025 |
|---------------------------------------------------------------------------|------------------------------------|

## Education

|                                                                       |                                 |
|-----------------------------------------------------------------------|---------------------------------|
| <b>PhD in Astrophysics</b><br>McMASTER UNIVERSITY                     | Hamilton, Canada<br>2021 - 2025 |
| <b>MSc in Astrophysics</b><br>McMASTER UNIVERSITY                     | Hamilton, Canada<br>2019 - 2021 |
| <b>BSc in Physics, with Minor in Mathematics</b><br>McGILL UNIVERSITY | Montreal, Canada<br>2016 - 2019 |

## Awards

|           |                                                                                                                                                          |                |
|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|
| 2026      | <b>IAU PhD Prize, Divison H (Interstellar Matter and Local Universe)</b><br>Awarded for the most impactful PhD thesis defended in 2025, in a given field |                |
| 2025      | <b>MPA Postdoctoral Fellowship</b><br>Max Planck Institute for Astrophysics, awarded for 2025 and postponed to 2027                                      |                |
| 2025-2027 | <b>NSERC Postdoctoral Fellowship</b><br>Natural Sciences and Engineering Research Council of Canada                                                      | \$140,000 CAD  |
| 2025      | <b>Governor General's Academic Gold Medal</b><br>McMaster University, awarded for highest academic standing at graduate level                            |                |
| 2025      | <b>Co-PI, ACCESS allocation PHY240335</b><br>U.S. National Science Foundation                                                                            | 2.6M CPU hours |
| 2023      | <b>Michael Smith Foreign Study Supplement, supporting research visit to MPA</b><br>Natural Sciences and Engineering Research Council of Canada           | \$6,000 CAD    |
| 2022-2025 | <b>Canada Graduate Scholarships – Doctoral</b><br>Natural Sciences and Engineering Research Council of Canada                                            | \$110,000 CAD  |
| 2021-2022 | <b>Queen Elizabeth II Scholarship in Science and Technology</b><br>Ontario Graduate Scholarship, Government of Ontario                                   | \$15,000 CAD   |

## Presentations

### CONFERENCE TALKS

Past five calendar years only. Invitation-only conferences are denoted by <sup>†</sup> and invited talks are denoted by <sup>††</sup>.

|                                                                                                                     |         |
|---------------------------------------------------------------------------------------------------------------------|---------|
| <i>Star Clusters as Guides to Galaxies</i> , IAU Symposium 411, San Carlos de Bariloche, Argentina                  | 11/2026 |
| <sup>††</sup> <i>Challenges and Innovations in Computational Astrophysics VII</i> , IAU Commission B1 (virtual)     | 11/2026 |
| <i>Journées Scientifiques Galaxies de l'ATCG 2026</i> , Paris, France                                               | 10/2026 |
| <i>MIAPbP workshop: Bridging Gaps in High-Mass Star Research</i> , Garching bei München, Germany                    | 09/2026 |
| <sup>†</sup> <i>Computational Galaxy Formation 2026</i> , Ringberg Castle, Tegernsee, Germany                       | 07/2026 |
| <sup>†</sup> <i>Exploring the Paths from High-z to Local Globular Clusters and Cluster Systems</i> , Sexten, Italy  | 06/2026 |
| <sup>†</sup> <i>Multiplicity Across Scales</i> , Heidelberg, Germany                                                | 04/2026 |
| <i>Ringberg MPA/MPE Postdoc Meeting</i> , Ringberg Castle, Tegernsee, Germany                                       | 02/2026 |
| <i>Bridging scales: star clusters and their host galaxies from the Local to the high-z Universe</i> , Matera, Italy | 09/2025 |
| <sup>†</sup> <i>Numerical Recipes in Star Formation</i> , Lorentz Center, Leiden, the Netherlands                   | 07/2025 |

|                                                                                                                        |         |
|------------------------------------------------------------------------------------------------------------------------|---------|
| <i>Globular Clusters: From the Milky Way to High-Redshift Galaxies</i> , Saint Mary's University, Halifax, Canada      | 06/2025 |
| <i>Annual CASCA Meeting</i> , Saint Mary's University, Halifax, Canada ( <b>runner-up, best talk</b> )                 | 06/2025 |
| <i>Star formation across the scales: star clusters to galactic disks</i> , McMaster University, Hamilton, Canada       | 08/2024 |
| <sup>†</sup> <i>2024 Alpbach workshop on clouds, star clusters &amp; black holes</i> , Alpbach, Austria                | 06/2024 |
| <i>Annual CASCA Meeting</i> , University of Toronto & York University, Toronto, Canada ( <b>runner-up, best talk</b> ) | 06/2024 |
| <i>Globular Clusters and Their Tidal Tails: From the Milky Way to the Local Group</i> , University of Toronto, Canada  | 05/2024 |
| <i>"Two in a million" - The interplay between binaries and star clusters</i> , ESO Headquarters, Garching, Germany     | 09/2023 |
| <i>MODEST-23: Star clusters in the post-pandemic era</i> , Northwestern University, Evanston, USA                      | 08/2023 |
| <i>Great Lakes Clusters and Streams</i> , University of Michigan, Ann Arbor, USA                                       | 08/2023 |
| <i>A multi-wavelength view on globular clusters near and far: from JWST to the ELT</i> , Sexten, Italy                 | 07/2023 |
| <i>Clusters @ McMaster Conference</i> , McMaster University, Hamilton, Canada                                          | 08/2022 |
| <i>TORCH workshop</i> , Center for Computational Astrophysics, New York, USA                                           | 08/2022 |
| <i>MODEST 21a Workshop</i> , Leiden University (virtual), Leiden, the Netherlands                                      | 07/2021 |
| <i>Annual CASCA Meeting</i> , University of Waterloo (virtual), Waterloo, Canada                                       | 06/2021 |

## INVITED SEMINARS & COLLOQUIA

|                                                                                                     |         |
|-----------------------------------------------------------------------------------------------------|---------|
| Institute seminar, Max Planck Institute for Astrophysics, Garching, Germany ( <b>colloquium</b> )   | 09/2026 |
| IAU Early-Career Summer Talk Series, International Astronomical Union (virtual, <b>colloquium</b> ) | 09/2026 |
| Munich University Observatory, Ludwig-Maximilian University, Munich, Germany                        | 05/2026 |
| Austrian Institute of Technology, Klosterneuburg, Austria                                           | 05/2026 |
| University of Vienna, Vienna, Austria ( <b>colloquium</b> )                                         | 05/2026 |
| Observatoire astronomique de Strasbourg, Strasbourg, France                                         | 06/2024 |
| Toronto Astrophysics Talks, University of Toronto, Toronto, Canada                                  | 03/2024 |
| University of Cologne, Cologne, Germany                                                             | 09/2023 |
| Galaxy Group Meeting, Max Planck Institute for Astrophysics, Garching, Germany                      | 06/2023 |
| Anton Pannekoek Institute, University of Amsterdam, Amsterdam, the Netherlands                      | 06/2023 |
| Institute of Astronomy, KU Leuven, Leuven, Belgium                                                  | 06/2023 |
| Seminar on Stellar Astrophysics, Max Planck Institute for Astrophysics, Garching, Germany           | 05/2023 |
| ESO Informal Discussion, ESO Headquarters, Garching, Germany                                        | 05/2023 |
| Pizza Lunch Informal Talk, Columbia University (virtual), New York, USA                             | 02/2021 |

## Professional experience

---

### STUDENT SUPERVISION

|                                                                                           |             |
|-------------------------------------------------------------------------------------------|-------------|
| Co-supervision of MSc students Isabella Armstrong and Zayna Khadour with Dr. Alison Sills | 2024 - 2026 |
| Co-supervision of MITACS student Edwin Laverde-Villarreal with Dr. Alison Sills           | 2024        |

### TEACHING

|           |                                                                                                                                                  |            |
|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------|------------|
| 2020-2025 | <b>Head Teaching Assistant: Introductory Physics for Engineering</b><br>Supervision of 55-60 TAs, guest lectures, development of marking schemes | 1500 hours |
| 2020-2022 | <b>Integrated Science Literature Review Course</b><br>Expert consultant, supervision of 5-6 students per semester                                | 50 hours   |
| 2019      | <b>Teaching Assistant: Introductory Physics for Engineering</b><br>Leading labs and tutorials, grading and invigilating exams; 35 students       | 65 hours   |
| 2019      | <b>Teaching Assistant: General Interest Astronomy</b><br>Leading tutorials, grading presentations; 35 students                                   | 65 hours   |

## Service

---

### SCIENTIFIC SERVICE

|                                                                                                           |           |
|-----------------------------------------------------------------------------------------------------------|-----------|
| Referee for Astronomy & Astrophysics                                                                      | 2024-now  |
| Member of hiring committee for an ERC-funded postdoctoral position, Max Planck Institute for Astrophysics | 2026      |
| Student Member, Hiring Committee for Observational Exoplanet Astronomy, McMaster University               | 2021-2022 |
| Student Member, Selection Committee for the Acting Associate Dean of Graduate Studies                     | 2020      |
| Organizer of the Stellar Astrophysics seminar series, Max Planck Institute for Astrophysics               | 2025-now  |
| LOC/SOC member, “Star formation across the scales: star clusters to galactic disks”, McMaster University  | 2024      |
| LOC/SOC member, Clusters @ McMaster Conference, McMaster University                                       | 2022      |
| Organizer of the summer colloquium series, Department of Physics & Astronomy, McMaster University         | 2022      |

### EQUITY, DIVERSITY & INCLUSION

|                                                                                                   |           |
|---------------------------------------------------------------------------------------------------|-----------|
| External Postdoc Representative, Max Planck Institute for Astrophysics                            | 2026-now  |
| Student Representative on Physics & Astronomy Faculty Meetings, McMaster University               | 2020-2024 |
| Student Member, Departmental Equity, Diversity and Inclusion (EDI) Committee, McMaster University | 2021-2022 |
| Mentor to seven first-year MSc students, McMaster University                                      | 2020-2025 |
| Active member & event organizer, Promoting Inclusion in Physics & Astronomy, McMaster University  | 2019-2025 |
| Communication Officer, STEM Support Committee, McGill University                                  | 2018-2019 |

### OUTREACH

|           |                                                                                                                                                                                                                                                                                                                                                                                                       |            |
|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|
| 2019-2025 | <b>Presenter for McMaster University’s on-campus planetarium</b><br>Presentations for school groups (5-18 years old) and the general public, including the shows <i>The Celestial Bear: The Six Nations’ Night Sky</i> and <i>Onekwá:tara – the Seven Dancers of the Pleiades</i> , developed in collaboration with Indigenous partners. Design and presentation of a themed show focused on nebulae. | 70+ shows  |
| 2022-2025 | <b>Presenter for McMaster University’s portable planetarium</b><br>Including eclipse-related shows and <i>The Celestial Bear: The Six Nations’ Night Sky</i> .                                                                                                                                                                                                                                        | 50+ shows  |
| 2024      | <b>Invited speaker, Royal Astronomical Society of Canada Toronto Centre</b><br>Online public talk for amateur astronomers on binary stars and star clusters.                                                                                                                                                                                                                                          |            |
| 2024      | <b>Eclipse ambassador for April 2024 total solar eclipse, McMaster University</b><br>Outreach to the general public, seven media interviews for CBC and local media.                                                                                                                                                                                                                                  |            |
| 2020      | <b>Invited presenter for Ask a Scientist</b><br>Online talk answering children’s questions on stars and planets.                                                                                                                                                                                                                                                                                      |            |
| 2016-2018 | <b>Guide, Parks Canada National Historic Sites in Quebec City</b><br>Guided tours and educational activities for school groups and the general public, in English and French. Development of guided tour scripts and training of new guides.                                                                                                                                                          | 150+ tours |

## Professional development

---

### TRAINING & CERTIFICATIONS

|                                                                                                                                                                                       |      |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| <b>Teaching and Learning Certificate of Completion, MacPherson Institute, McMaster University</b><br>Training in teaching methods and course development, three semester-long courses | 2022 |
| <b>Professor Hippo-on-Campus Certificate of Completion, McMaster University</b><br>Student mental health education program for educators                                              | 2022 |
| <b>Mental Health First Aid, Mental Health Commission of Canada</b><br>Training course in how to assist someone experiencing a mental health crisis                                    | 2019 |